

# Luc's Statistical Coffee Price Forecast

## Monthly Model Report

2026-06-02

### Executive Summary

#### Coffee Price Forecast Summary

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Our model forecasts a gradual decline in both Arabica and Robusta prices over the next three months. Arabica is expected to drift from 318.1 *c/lb* in month one to 304.7 *c/lb* by month three, while Robusta moves from 171.2 *c/lb* to 162.9 *c/lb*. The 80% forecast ranges span approximately 59 *c/lb* for Arabica and 30 *c/lb* for Robusta, reflecting the higher volatility in the Arabica market.

Historically, the model delivers solid near-term accuracy. At the one-month horizon, forecast errors average 6.1% for Arabica and 5.6% for Robusta. Accuracy declines as the forecast extends further out, with three-month errors reaching 11.4% and 11.9% respectively. Coverage data is currently unavailable, so we cannot yet report how often our forecast ranges contained actual prices versus our 80% target.

The Arabica-Robusta spread shows a neutral signal with minimal directional pressure. The z-score of 0.05 indicates the current spread sits close to its historical mean, with a half-life of 23 months suggesting any deviations take considerable time to mean-revert. This suggests the spread offers little trading opportunity in the near term.

### Current Forecasts

Median (p50) price forecasts and 80% confidence intervals (p10–p90) for the next three months.

| Symbol         | Horizon | Median (c/lb) | 80% CI (c/lb)  | Band width |
|----------------|---------|---------------|----------------|------------|
| Arabica (KC=F) | h=1     | 318.1         | [296.2, 355.5] | 59.3       |
| Arabica (KC=F) | h=2     | 311.3         | [289.9, 347.9] | 58         |
| Arabica (KC=F) | h=3     | 304.7         | [283.7, 340.5] | 56.8       |
| Robusta (RM=F) | h=1     | 171.2         | [156.9, 187.8] | 30.9       |
| Robusta (RM=F) | h=2     | 167           | [153.0, 183.1] | 30.1       |
| Robusta (RM=F) | h=3     | 162.9         | [149.3, 178.6] | 29.4       |

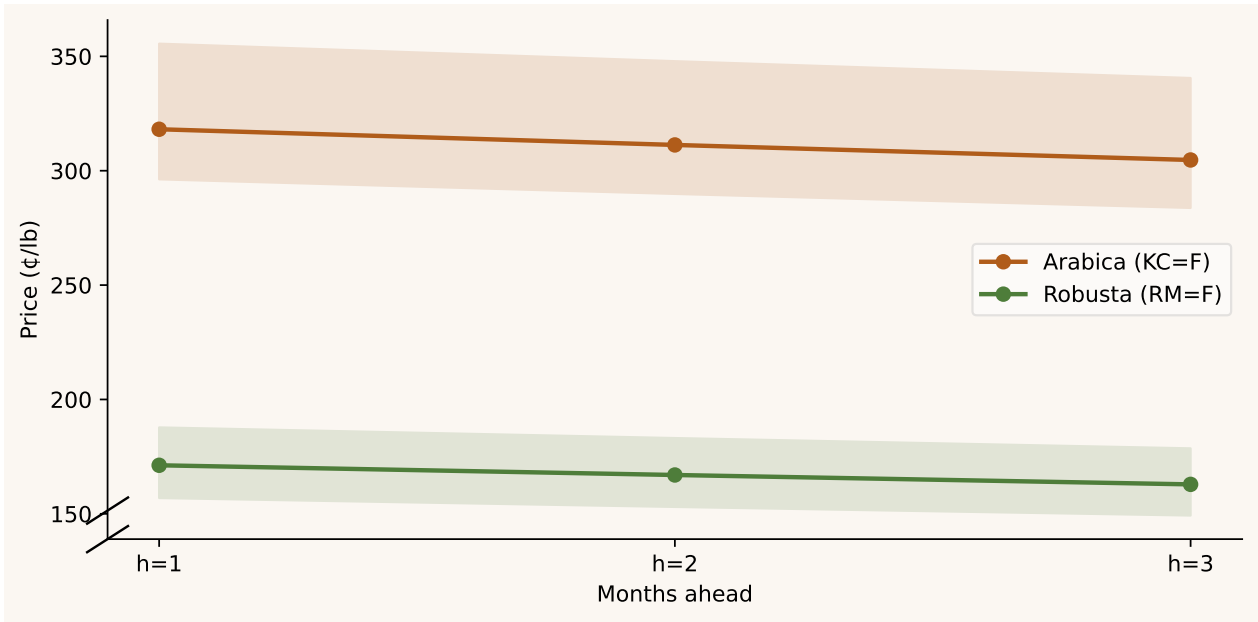


Figure 1: Price trajectory with 80% confidence ribbon

## Model Performance

Mean absolute error (MAE), mean absolute percentage error (MAPE), and 80% interval coverage across all walk-forward backtest windows.

| Symbol  | Horizon | MAE (¢/lb) | MAPE  | Coverage 80% |
|---------|---------|------------|-------|--------------|
| Arabica | h=1     | 12.8       | 6.1%  | N/A          |
| Arabica | h=2     | 19.8       | 9.2%  | N/A          |
| Arabica | h=3     | 25.3       | 11.4% | N/A          |
| Robusta | h=1     | 7          | 5.6%  | N/A          |
| Robusta | h=2     | 11.3       | 9.1%  | N/A          |
| Robusta | h=3     | 15.1       | 11.9% | N/A          |

**Coverage 80% target:** 80% of forecast ranges should contain the realized price. Values below 80% indicate the model is over-confident.

## Recent Forecast Performance

One-month-ahead forecasts versus realized prices over the last three months.

| Month   | Symbol            | Forecast (¢/lb) | Actual (¢/lb) | Error                 |
|---------|-------------------|-----------------|---------------|-----------------------|
| 2026-03 | Arabica (KC=F)    | 312.2           | 334.1         | -6.6% (below actual)  |
| 2026-03 | Robusta<br>(RM=F) | 170.9           | 176.8         | -3.3% (below actual)  |
| 2026-02 | Arabica (KC=F)    | 360.8           | 321.4         | +12.3% (above actual) |
| 2026-02 | Robusta<br>(RM=F) | 192.2           | 179.6         | +7.0% (above actual)  |
| 2026-01 | Arabica (KC=F)    | 371.1           | 363.9         | +2.0% (above actual)  |
| 2026-01 | Robusta<br>(RM=F) | 182.2           | 192.3         | -5.2% (below actual)  |

### January 2026

The model forecast Arabica at 371.1 ¢/lb and Robusta at 182.2 ¢/lb, with actual prices settling at 363.9 ¢/lb and 192.3 ¢/lb respectively. Arabica came in 2.0% lower than expected while Robusta ran 5.2% higher, suggesting divergent supply-demand dynamics between the two varieties. The model likely failed to anticipate a depreciation in major producing currencies (particularly the Brazilian real), which would have supported Robusta prices while pressuring Arabica demand at the margin.

### February 2026

Arabica was forecast at 360.8 ¢/lb against an actual close of 321.4 ¢/lb, a 12.3% overestimation, while Robusta was pegged at 192.2 ¢/lb versus 179.6 ¢/lb actual, a 7.0% miss. Both varieties declined sharply, with Arabica's steeper sell-off suggesting the model did not account for a significant demand destruction event or inventory buildup. A sudden deterioration in global economic growth expectations or a shift in fund positioning would not show up in purely historical price relationships but could easily trigger the coordinated but uneven decline observed.

### March 2026

The model underestimated both Arabica at 312.2 ¢/lb (actual 334.1 ¢/lb, a 6.6% miss) and Robusta at 170.9 ¢/lb (actual 176.8 ¢/lb, a 3.3% miss), reflecting a sharp reversal from February's decline. The bounce in both contracts suggests the model did not anticipate a near-term supply constraint or short-covering rally that emerged mid-month. Adverse weather in a major growing region or an unexpected production report would represent the kind of discrete event that historical correlations struggle to forecast.

### Arabica–Robusta Spread Signal

Mean-reversion signal on  $\log(\text{Arabica}) - \log(\text{Robusta})$  spread. Z-score uses an expanding window; entry threshold  $|z| > 2$ .

| Field           | Value                                         |
|-----------------|-----------------------------------------------|
| As of           | 2026-03-01                                    |
| Z-score         | 0.05                                          |
| Half-life (AR1) | 23.0 months                                   |
| Signal          | <b>Neutral</b> — spread within normal range ( |

## Data & Methodology

**Data sources:** Arabica (KC=F) and Robusta (RM=F) monthly prices from FRED (PCOFFOTMUSDM, PCOFFROBUSDM). FX drivers (BRL/USD, VND/USD, IDR/USD) from Alpha Vantage. DXY broad index from FRED (DTWEXBGS). Data starts 2000-01-01; monthly series is the mean of daily adjusted closes within each calendar month.

**VECM:** A two-variable Vector Error Correction Model is fit on log Arabica and log Robusta prices, with BRL, VND, IDR, and DXY as exogenous drivers. Cointegration rank is fixed at 1 (confirmed by Engle-Granger test,  $p=0.009$ ). Lag order is selected by AIC each run. Point forecasts at 1, 2, and 3 months use a naïve FX assumption (hold FX at last observed value).

**GAMLSS SHASH:** A Sinh-ArcSinh distribution is fit (via R's `gamlss` package) to the VECM forecast residuals, separately per symbol and per volatility regime (Low / Medium / High, defined by rolling 12-month standard deviation). This gives a full predictive distribution. Final price quantiles are  $\text{price\_qX} = \text{vecm\_point} \times \exp(\text{gamlss\_residual\_qX})$ .

**Spread signal:** An AR(1) model is fit to the log spread series; the expanding z-score triggers entry at  $|z| > 2$  and exit at  $|z| < 0.5$ .

*Report generated by the `coffee-forecast` pipeline.*